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Next item →

Learning Check - Additional Probability

Review Learning Objectives

Instructions

coach

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Assignment details

Due Oct 2, 11:59 PM IST

Attempts 4 left (5 attempts every 24 hours)

Try again

Submitted Sep 27, 1:19 AM IST

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1. We saw in the lectures that for any random variable X and constants a, b we have $\mathbb{E}[aX+b] = a\mathbb{E}[X]+b$. Which of the following for any random variable X is true? **1 / 1 point**

- ○ $\mathbb{E}[\frac{1}{X}] = \frac{1}{\mathbb{E}[X]}$
- ○ $\mathbb{E}[X^2] = \mathbb{E}[X]^2$
- ○ $\mathbb{E}[X^{\frac{1}{3}}] = \mathbb{E}[X]^{\frac{1}{3}}$
- ◉ None of the above are true (in fact, for any non-linear function $f(x)$, we don't necessarily have $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$)

Correct

2. Consider two random variables X, Y with joint pdf $f_{X,Y}(x,y)$. Which of the following statements is ***not*** correct? **1 / 1 point**

- ○ The pdf of X, denoted by $f_X(x)$, can be obtained by integrating $f_{X,Y}(x,y)$ in the variable y; i.e. $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy$
- ○ $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)dxdy = 1$
- ◉ $f_{X,Y}(x,y) = f_X(x)f_Y(y)$, where f_X and f_Y are the pdf's of X, Y, respectively
- ○ All the above are correct.

Correct
This is not generally true.

3. Let X, Y be two independent random variables. **1 / 1 point**

We know that for arbitrary functions f, g we have $\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)]$. Which of the following is not true in general?

- ○ $\mathbb{E}[XY+4] = \mathbb{E}[X]\mathbb{E}[Y]+4$
- ◉ $\mathbb{E}[(XY)^2] = \mathbb{E}[XY]^2$
- ○ $\mathbb{E}[X^3Y] = \mathbb{E}[X^3]\mathbb{E}[Y]$
- ○ All the above are true.

Correct
This equation is of the form $E[f(X,Y)]$ so we cannot apply the rule from above.